TECHNICAL EVALUATION & PRESENTATION GUIDE

LegalAI System Overview

An Enterprise-Grade, Full-Stack Artificial Intelligence Legal Advisory Platform & Judicial Management Ecosystem for India

Architecture: 4-Tier Distributed System | **Database:** SQLite 3 (WAL Mode) | **Intelligence:** Hybrid-RAG (BM25 + FAISS + LLM) | **Jurisdiction:** India (IPC / BNS 2023)

🎯 1. Executive Summary & Problem-Solution Fit

⚠️ The Real-World Legal Problem in India

1. Historic Legal Transition: India is actively transitioning from colonial penal codes (IPC 1860, CrPC 1973, Evidence Act) to modern laws (Bharatiya Nyaya Sanhita - BNS 2023, BNSS, BSA). Generic LLMs hallucinate outdated sections.
2. High Cost & Delay: Ordinary citizens struggle to understand legal procedures, find the right jurisdiction court, or connect with verified advocates without exploitation.

✅ The LegalAI Solution

1. Grounded Hybrid RAG: Combines lexical BM25 with FAISS dense vector search to provide zero-hallucination statutory legal citations.
2. BNS Temporal Transition Engine: Automatically maps old IPC sections to new BNS 2023 sections.
3. Direct Judicial Integration: Instantly connects user case queries to specific courts (Apex to Taluka) and verified State Bar advocates.

📐 2. High-Level 4-Tier System Architecture

LegalAI is engineered as a decoupled, microservice-ready system providing high speed, strict data isolation, and graceful fallback mechanisms:

TIER 1: CLIENT FRONTEND (Single Page Application) — Port: Web Client

HTML5 · CSS3 (Custom Luxury Dark Gold Aesthetic) · Vanilla ES6+ JS · Zero Runtime Bloat
Landing Page (index.html) · Advocate Directory (lawyers.html) · Client Portal (user.html) · Admin (admin.html)

▼ REST API / HTTPS JSON Calls ▼

TIER 2: API GATEWAY & APPLICATION SERVER — Port: 5000 (Node.js & Express)

JWT Bearer Authentication · Bcrypt Hashing · Rate Limiting · Private Network Access CORS
Routes: /api/auth · /api/ai/chat · /api/lawyers · /api/courts · Dataset Watcher & Sync Service

▼ High-Speed Inter-Process & HTTP REST ▼

TIER 3: INTELLIGENCE & HYBRID RAG MICROSERVICE — Port: 8000 (Python FastAPI)

Calibrated LinearSVC Domain Classifier · Dense FAISS Cosine Vector Space · BM25 Exact Lexical Search
Cross-Encoder Reranker · IPC vs BNS 2023 Transition Engine · Cloud Fallback: Google Gemini 1.5

▼ ACID Relational Transactions & Disk Synchronization ▼

TIER 4: RELATIONAL DATABASE & PERSISTENCE — better-sqlite3 (WAL Mode)

SQLite DB (server/legalai.db) · Tables: users, lawyers, courts, consultations, ai_chats
Flat Datasets: Real-time bi-directional synchronization with datasets/ & ml/dataset/ (JSON & CSV)

Key Architectural Advantage: If the local Python ML server is unreachable, the Node.js backend automatically falls back to an integrated domain rulebook and Cloud Gemini API. The system never crashes or shows a dead screen.

🧩 3. The 6 Core Modules Explained Simply (Part I)

🔐 Module 1: User Authentication & Role-Based Access Control (RBAC)

What It Does: Manages secure user registration, password encryption, sign-in, and permission scopes.
Technical Details:

- **Password Security:** Bcrypt hashing with 10 salt rounds. Plaintext passwords are never saved.
- **Session Integrity:** Stateless JSON Web Tokens (JWT) signed with HMAC-SHA256, carrying user role and ID.
- **Strict Form Validation:** Full Name, Email, Mobile Number (10–15 digits verified), and Password confirmation matching.
- **Role Isolation:** Enforces boundaries between regular Clients (`user`), Advocates (`lawyer`), and Administrators (`admin`).

🤖 Module 2: AI Legal Assistant & Hybrid-RAG Intelligence Engine

What It Does: Provides interactive legal answers, statutory citations, court jurisdiction guidance, and procedural checklists.
Technical Pipeline:

1. **Domain Intent Classifier:** Calibrated LinearSVC classifies queries into categories (Criminal, Family, Corporate, Property, etc.).
2. **Sparse + Dense Hybrid Retrieval:**
   - *BM25:* Catches exact statutory numbers (e.g. "Section 138 NI Act").
   - *FAISS:* Catches natural language semantic meaning in sub-millisecond vector cosine space.
3. **Cross-Encoder Reranker:** Re-scores top-K results to eliminate irrelevant matches.
4. **IPC 1860 vs. BNS 2023 Transition Engine:** If user asks about an old IPC section (e.g., IPC 302 or 420), it dynamically attaches the new BNS 2023 equivalent (BNS § 103 or § 318(4)).
5. **Multi-Turn Memory:** Saves conversation history in `ai_chats` table to understand contextual follow-up questions.

⚖️ Module 3: Verified Advocate & Senior Counsel Directory

What It Does: Comprehensive registry of 50+ verified Indian advocates, senior counsel, and law firm partners.
Technical Highlights:

- **7-Axis Faceted Search:** Instant search by Name/Bio, Practice Specialty, Practicing Court Code, Fee Ceiling, Rating (1.0–5.0), Availability, and City.
- **Bar Verification:** Includes official Bar Council enrollment numbers (e.g., D/248/1980) and verifiable courtroom win rates.
- **Interconnected Judicial Mapping:** Advocates are programmatically linked to courts via `courtCodes` (e.g., `SCI`, `DHC`, `NCLAT`).

🧩 3. The 6 Core Modules Explained Simply (Part II)

🗓️ Module 4: Consultation Booking & Collision-Free Scheduling System

What It Does: Connects clients directly to specialist advocates with automated appointment booking and scheduling.
Step-by-Step Workflow:

1. **Authentication Gate:** When user clicks "Book Consultation", system checks JWT token. If not logged in, it prompts Login/Register and remembers the booking intent.
2. **Auto-Resumption:** Immediately upon successful authentication, the booking form reopens pre-filled with the user's name, email, and phone.
3. **Details Collection:** Client picks Date (min = today), Time Slot (e.g. 11:30 AM - 12:30 PM), Lawyer, and enters a brief Case Summary.
4. **Duplicate & Collision Check:** The backend checks if the client or lawyer already has an active booking on that date and slot.
5. **Unique Booking ID:** Generates unique cryptographic tracking code (e.g. `BK-LGL-849201`).
6. **Management Portal:** User can view active vs. past bookings, copy booking ID, or cancel/restore sessions in "My Consultations".

🏛️ Module 5: Courts Directory with GIS & Live Map Navigation

What It Does: Directory of 100+ Indian judicial complexes from Supreme Court down to Taluka Magistrate courts.
Technical Details:

- **Judicial Hierarchy:** Categorized by Apex Court (`SCI`), State High Courts (Delhi, Bombay, Karnataka, Allahabad, etc.), Central Tribunals (`NCLAT`, `NGT`), and District/Sessions courts.
- **GIS Navigation:** Real GPS latitude/longitude coordinates, direct Google Maps turn-by-turn routing, and live embedded maps.
- **Operational Metadata:** Official e-Courts CIS filing systems, clerk counter hours, registrar contacts, and appellate circuit benches.

🔄 Module 6: Real-Time Data Sync & File-System Watcher

What It Does: Eliminates data drift between data scientists working on CSV/JSON datasets and web clients using SQLite.
Technical Details:

- **Two-Way Synchronization:** Edits in the SQLite database automatically write back to `datasets/courts_dataset.csv` and `lawyers_dataset.json`.
- **Automated Disk Watcher (`fs.watch`):** Modifying external CSV/JSON files automatically updates the database within 500ms using atomic SQL `UPSERT` transactions.
- **Fault-Tolerant File Locking:** Handles Windows file locks (`EBUSY`) with retry-backoff and temporary file staging.

🔬 4. Key Technical Innovations & Evaluation Highlights

1. Why Hybrid RAG (BM25 + FAISS)?

• **The Problem:** Dense vector embeddings often confuse nearby numbers (e.g. "Section 138 NI Act" vs "Section 139").
• **The Solution:** BM25 guarantees 100% precision on exact section numbers, while FAISS captures conceptual semantic meaning. Combining them yields near-zero statutory hallucination.

2. Why SQLite WAL Mode?

• **The Problem:** Traditional external SQL databases (MySQL/PostgreSQL) introduce network latency and heavy runtime dependencies.
• **The Solution:** SQLite with `better-sqlite3` runs in-process with sub-millisecond execution. WAL (Write-Ahead Logging) allows concurrent readers without write blocking.

3. Temporal Legal Validity (BNS Engine)

• **The Innovation:** Dual-citation pipeline automatically cross-references old Indian Penal Code sections with new Bharatiya Nyaya Sanhita (2023) provisions.
• Eliminates outdated legal advice in post-July 2024 Indian judicial proceedings.

4. Resilient Multi-Tier Fallback

• **Tier A:** Local Python RAG microservice on port 8000.
• **Tier B:** Node.js statutory domain catalog with pattern matching.
• **Tier C:** Google Gemini 1.5 Cloud LLM API.
• Zero single point of failure — service availability is always guaranteed.

📊 5. Database Schema Matrix

Table Name	Primary Key	Key Fields & Indexes	Relational Purpose
users	`id` (Auto-inc)	`email` (UNIQUE), phone, `password` (Bcrypt), `role`	User authentication, client profiles & role-based permissions
lawyers	`id` (Auto-inc)	`barNumber` (UNIQUE), `specialty`, `courtCodes`, `rating`	Advocate registry, bar license verification, fee schedule
courts	`id` (Auto-inc)	`code` (UNIQUE), `jurisdiction`, `level`, `latitude`, `longitude`	Apex, High, District & Taluka courts, GIS mapping coordinates
consultations	`id` (Auto-inc)	`bookingId` (UNIQUE), `lawyerId` (FK), `userId` (FK), `status`	Booked client appointments, scheduled dates, collision checks
ai_chats	`id` (Auto-inc)	`userId` (FK), `role`, `category`, `timestamp`	Multi-turn context persistence for follow-up legal queries

🎤 6. Step-by-Step 3-Minute Live Presentation Script

Minute 1: The Problem & The AI Legal Assistant (Demo)

"Evaluators, India is currently undergoing a legal revolution transitioning from IPC 1860 to BNS 2023. Notice what happens when I ask our LegalAI Assistant: 'What is the punishment for murder under new criminal laws?' It doesn't hallucinate. It immediately cites BNS 2023 Section 103, compares it with former IPC 302, cites procedural bail under BNSS § 483, and recommends advocates practicing before the Sessions Court."

Minute 2: The Courts GIS Directory & Advocate Matching

"Next, law does not exist in a vacuum — it requires courts and advocates. Under our Courts Directory, we can view 100+ courts from the Supreme Court of India down to District Courts. Clicking any court opens live Google Maps navigation and displays verified advocates licensed before that specific court bench."

Minute 3: The Consultation Booking Flow with Auth-Gate

"Finally, when a user clicks 'Book Consultation', our system intercepts the request. If unauthenticated, it prompts our secure sign-in modal. Upon authentication, it instantly reopens the booking dialogue pre-filled with the user's name, email, and mobile. Once submitted, it verifies that there is no time-slot collision, generates a unique Booking ID (e.g. BK-LGL-849201), and syncs with SQLite and our dataset files."

❓ 7. Viva & Technical Panel Q&A Cheat-Sheet

Q1: Why did you choose SQLite over PostgreSQL or MySQL?

Answer: SQLite with Write-Ahead Logging (WAL) runs directly inside the Node.js process using C++ bindings (`better-sqlite3`). This provides microsecond query times with zero TCP network latency, zero server configuration overhead, and seamless integration with our live CSV/JSON synchronization service.

Q2: How do you prevent data loss or duplicate bookings?

Answer: We enforce duplicate checks at both the database schema level (UNIQUE constraints on `bookingId`) and the API layer: a SQL query verifies that neither the client nor the advocate has an existing non-cancelled booking on that date and time slot before committing the transaction.

Q3: How is client privacy maintained?

Answer: We utilize cryptographic JWT Bearer tokens. All consultation retrieval routes extract the user ID directly from the validated token signature, ensuring users can only ever access their own consultations. Passwords are never saved in plain text.

LegalAI Project Status: Fully Operational · 100% Test Suite Passed · Production Ready.